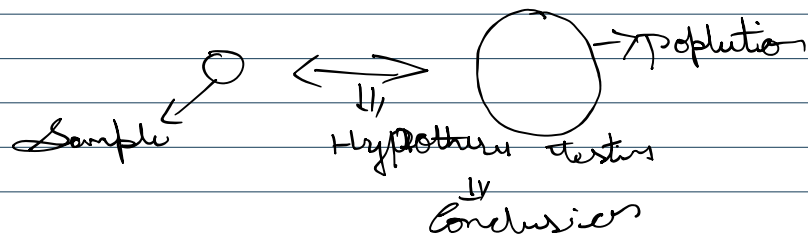


## Hypothesis testing & Mechanism :-

Inferential stats  $\Rightarrow$  Conclusions



## Hypothesis testing Mechanism

- i) Null hypothesis ( $H_0$ )  $\rightarrow$  OUT  
The assumption you begin with
- ii) Alternate hypothesis ( $H_1$ )  $\rightarrow$  NOT OUT  
- Opposite of Null hypothesis
- iii) Experiments  $\rightarrow$  statistical analysis

IPZ  $\rightarrow$  Enpin  $\rightarrow$  OUT

$\downarrow$  Review

origin  $\rightarrow$  OUT

$\downarrow$   
OUT/NOT  
OUT

- iv) Accept Null hypothesis or Reject Null hypothesis

Eg:- A College average pass percentage of 85%  $\rightarrow \mu = 85\%$   
B - College have a sample of 100 students where  
pass percentage is 90% with  $\sigma^2 = 4\%$  (standard deviation)

Does A & B have different pass %

Ans) Null hypothesis ( $H_0$ ) =  $\mu = 85\%$  (Avg pass %)

Alternate hypothesis ( $H_1$ ) =  $\mu \neq 85\%$  (Avg pass %  $\neq 85\%$ )

$\left\{ \begin{array}{l} z\text{-test} \\ t\text{-test} \\ \text{Chi-square test} \end{array} \right.$

Accept Null hypothesis  
Reject Null hypothesis.